

Light-flavour backgrounds and photon veto proposals

v3 Stage 2; fixed 1,091-chunk BDT; full 1,200-chunk Z_{cc}/Z_{ss} campaigns

FCC-ee $\Lambda_b \rightarrow \Lambda \gamma$ analysis

4 October 2026

Question and frozen samples

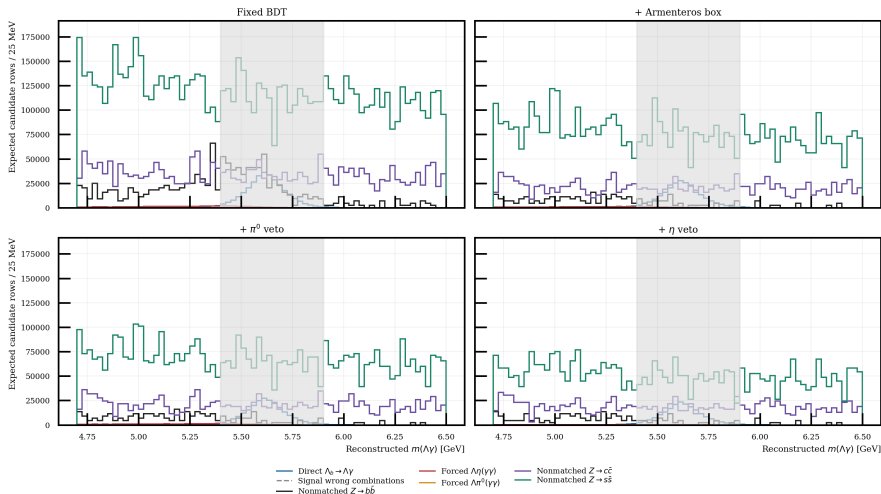
- Compare direct signal, nonmatched $Z_{bb}/Z_{cc}/Z_{ss}$, and forced $\Lambda\eta/\Lambda\pi^0$ candidates at fixed BDT score ≥ 0.9538269639 .
- Z_{bb} : 1,091 v3 chunks, 395,623,929 processed input events. Z_{cc} : 1,200 chunks, 499,786,495 events. Z_{ss} : 1,200 chunks, 499,842,440 events.
- Direct Physics signal: 100,002 generated direct decays. Forced η and π^0 : 100,000 and 100,005 generated direct decays. Both charge conjugates included.
- Plot physically weighted *candidate rows*, with 5.4–5.9 GeV as provisional reconstructed peak. Truth labels are diagnostic after candidate construction.

Normalization and paired proposal

- $N_Z = 6 \times 10^{12}$; $\mathcal{B}(Z \rightarrow b\bar{b}) = 0.15$, $\mathcal{B}(Z \rightarrow c\bar{c}) = 0.1203$, $\mathcal{B}(Z \rightarrow s\bar{s}) = 0.156$ as a down-type flavour-equality scenario.
- Each inclusive weight is $N_Z \mathcal{B}(Z \rightarrow q\bar{q}) / N_{\text{processed},q}$, applied to nonmatched candidate rows. Forced modes stay separate from inclusive Zbb .
- Four paired stages: fixed BDT; add Armenteros box $0.67 \leq |\alpha| \leq 0.78$, $0.075 \leq q_T \leq 0.120$ GeV; add π^0 photon-pair veto ± 20 MeV; add η veto ± 50 MeV.
- All stages use reconstructed quantities only. Windows and score remain proposals pending PI review and angular response work.

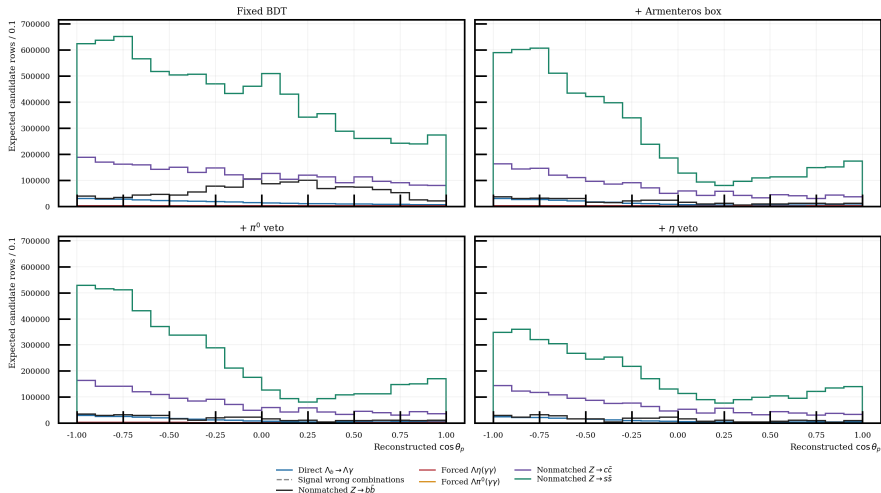
Expected reconstructed mass: all components, linear scale

All components: linear expected-candidate scale



Expected reconstructed $\cos \theta_p$: all components, linear scale

All components: linear expected-candidate scale



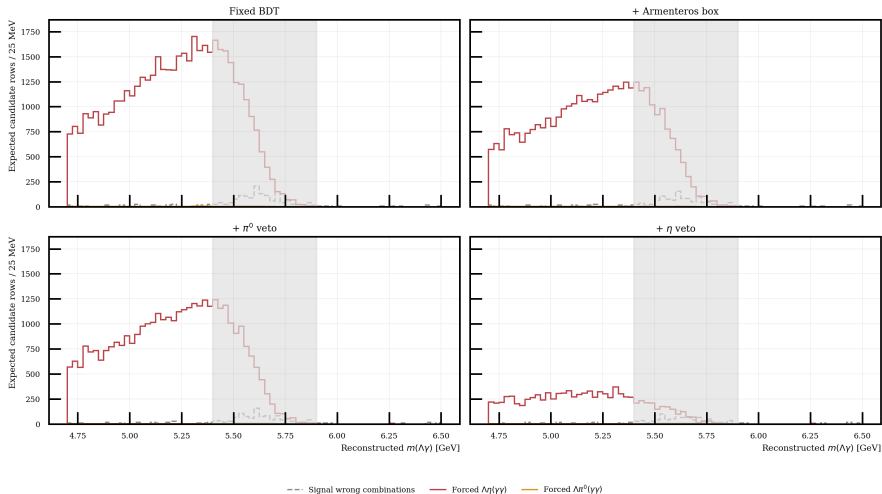
Peak expected candidates (5.4–5.9 GeV)

Step	Signal	Zbb	Zcc	Zss	Forced η	Forced π^0
BDT	301,335	550,523	677,338	2,286,433	13,136	107
+ Armenteros	231,473	79,621	410,158	1,486,837	9,844	84
+ π^0 veto	228,827	79,621	410,158	1,308,940	9,785	66
+ η veto	194,757	61,422	363,943	962,511	1,924	60

Zcc peak rows: 469 $\rightarrow$ 252; Zss: 1,221 $\rightarrow$ 514. Final candidate-count MC 95% intervals: Zcc 320,391–411,762; Zss 881,089–1,049,433; Zbb 40,477–89,366. These omit normalization and detector systematics.

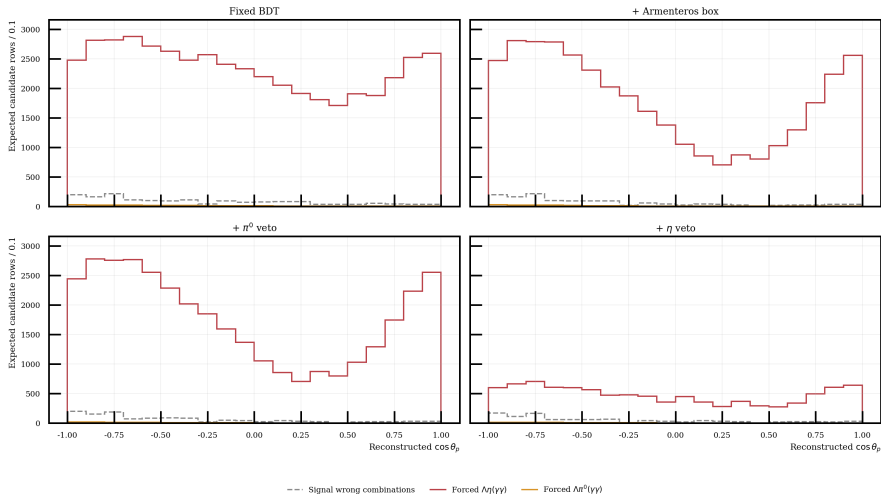
Smaller components: mass, linear scale

Feed-down / light-flavour detail: linear expected-candidate scale



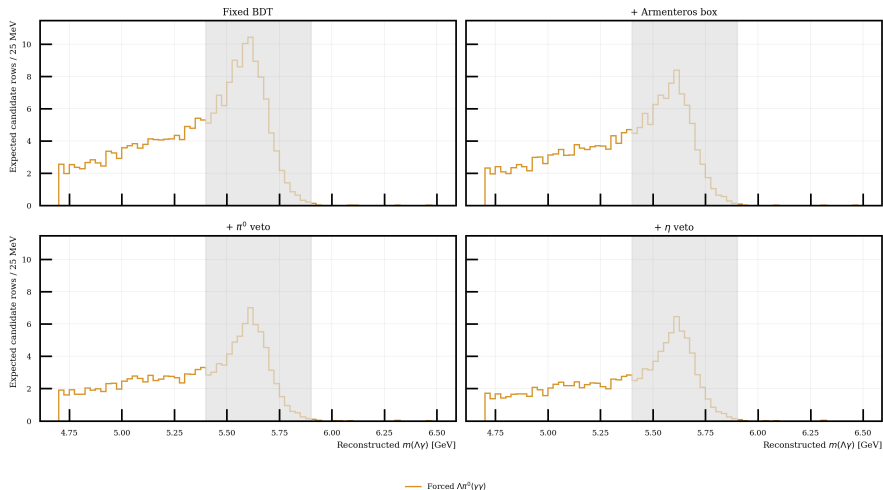
Smaller components: $\cos\theta_p$, linear scale

Feed-down / light-flavour detail: linear expected-candidate scale

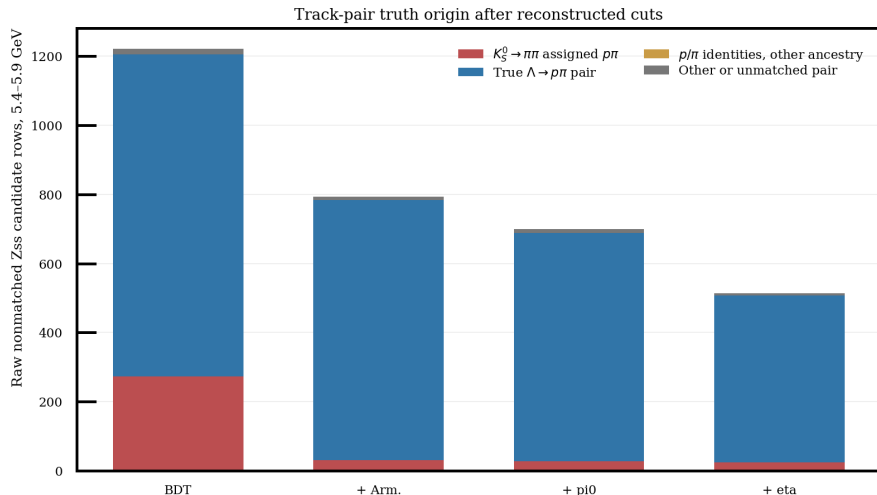


Forced π^0 mass detail, linear scale

$\Lambda\pi^0$ detail: linear expected-candidate scale

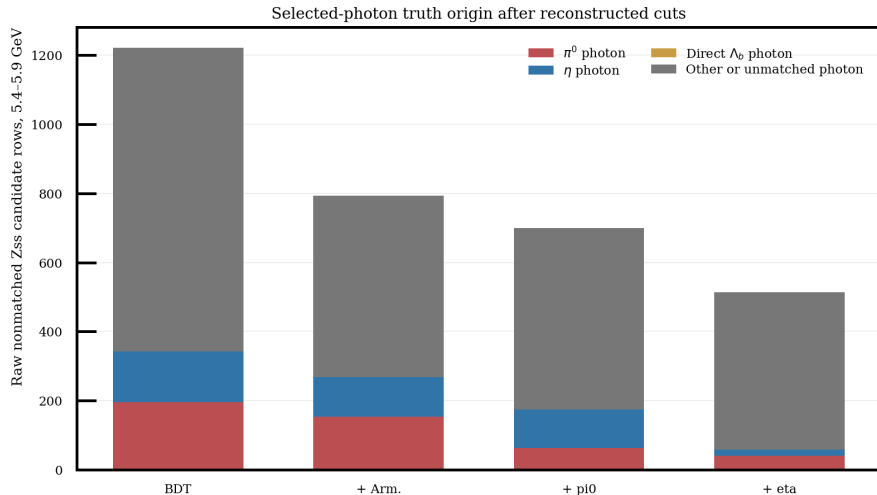


Zss truth: a real Λ pair dominates



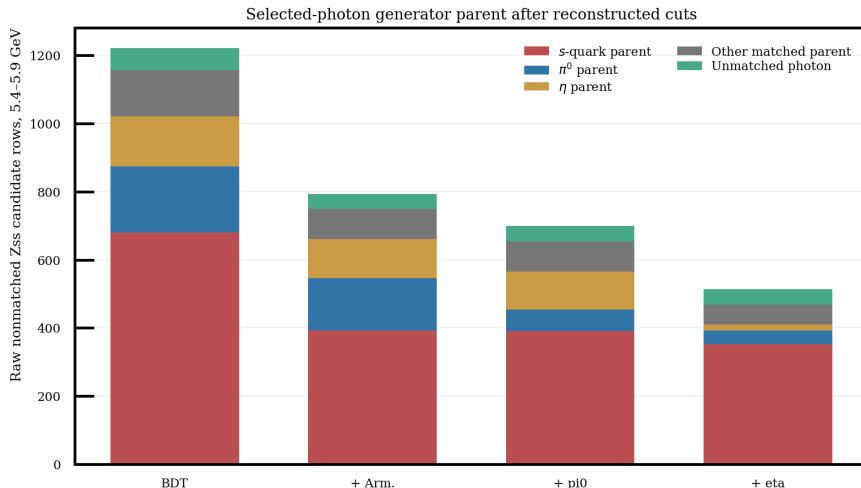
The Armenteros box removes 242/273 true K_S^0 pairs, but 752/932 true Λ background pairs survive. Direct signal retention is 76.8%.

Photon origins in Z_{ss}



After all proposals, 429/514 peak candidates combine a true Λ pair with a photon outside the π^0/η -parent categories. Parent-quark links need generator-record interpretation.

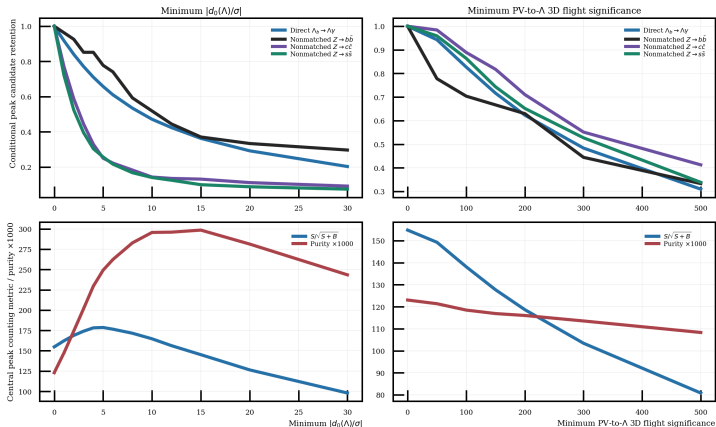
Tracing the selected photon through the stored MC chain



Final peak: 352/514 photons have a recorded s-quark parent; 347 of those reach a Z within four stored ancestor generations. The first-parent chain supports a strange-parton origin, while the full event graph is needed to name the shower mechanism.

Follow-up: reconstructed Λ displacement

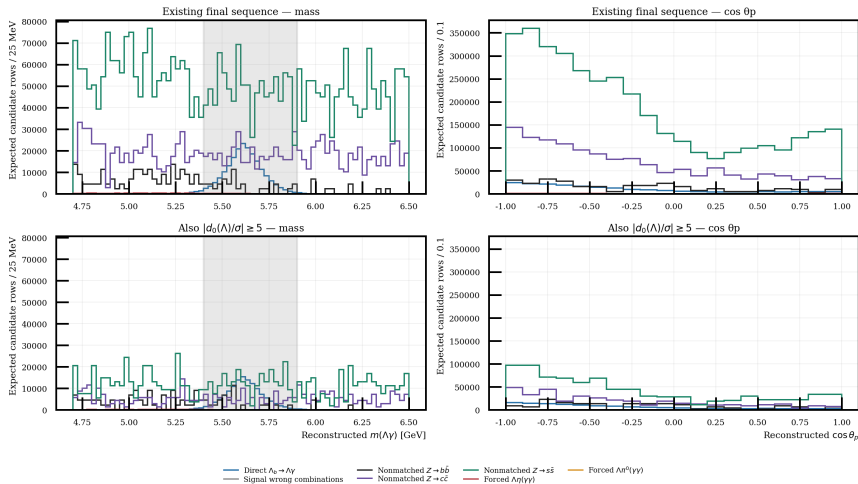
After fixed BDT + Armenteros + π^0 + η proposals; physically weighted inclusive backgrounds



Paired scan after BDT + Armenteros + π^0 + η proposals. $|d_0(\Lambda)/\sigma| \geq 5$ retains 65.8% signal and 25.7% Zss peak rows; PV-to-SV flight significance adds little.

Paired mass and angle: illustrative $|d_0/\sigma| \geq 5$

Paired linear-scale expected candidate distributions



Interpretation and PI decision

- Zss is the largest projected peak component under the stated branching scenario. An eventual weighted BDT mixture should follow dedicated Zss ancestry and independent test studies.
- The π^0 veto retains 98.9% of direct signal after Armenteros and 88.0% of Zss. The subsequent η veto retains 85.1% of signal and 73.5% of Zss while rejecting 80.3% of forced η .
- $S/(S + B_{bb} + B_{cc} + B_{ss})$ rises from 7.9% at fixed BDT to 12.3% after all proposals. This excludes forced modes to avoid double counting and is not a fit purity.
- Decide whether to optimize veto windows and repeat PHSP angular acceptance and resolution checks. No score or veto is adopted by this deck.
- The additional Λ impact-parameter threshold is exploratory. Validate independently, check PV/SV resolution and recompute PHSP angular acceptance before adopting it.